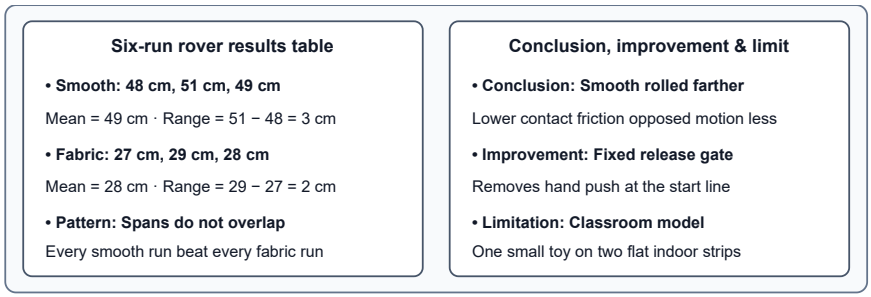
Reference sheets

Reference A · Six-run rover results table, means, ranges, and friction

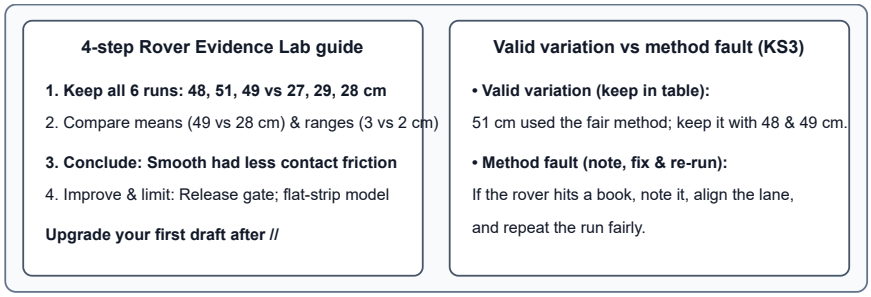


Six-run rover results table. It shows smooth and fabric runs, means, ranges, and friction.

Reference only. Do not write on this sheet. Use all six values to support your conclusion.

Surface condition.	Run 1, Run 2, Run 3 (cm).	Mean and range (cm).	Key Stage 3 science reason and limit.
1. Smooth surface.	48 cm, 51 cm, 49 cm (keep all three).	Mean = 49 cm. Range = 51 − 48 = 3 cm.	Lower contact friction opposed rover motion less on smooth surface.
2. Fabric surface.	27 cm, 29 cm, 28 cm (keep all three).	Mean = 28 cm. Range = 29 − 27 = 2 cm.	Greater contact friction opposed rover motion more on fabric.
3. Pattern across conditions.	Smooth span (48–51 cm) vs Fabric span (27–29 cm).	Difference between means = 21 cm.	Spans do not overlap. Limit: tested one small rover on two flat strips.

Reference B · Four-step Rover Evidence Lab frame and // upgrade



Four-step Rover Evidence Lab guide. It shows keeping six runs, comparing means, concluding, and upgrading with //.

Lab step.	Sentence stem for your book.	What to check before and after //.
1. Evidence (Keep all 6 runs).	Smooth runs were ___ cm (mean ___ cm, range ___ cm). Fabric runs were ___ cm (mean ___ cm, range ___ cm).	Did you keep all three repeats for each surface with cm units?

Lab step.	Sentence stem for your book.	What to check before and after //.
2. Conclusion (Link friction).	In this test, the rover rolled farther on ___ because ___ created less contact friction.	Did you name contact friction and bound the claim to this test?
3. Improvement & limitation.	To improve the method, I would add a ___. One model limitation is ___.	Did you distinguish a method improvement from a model limitation?
4. Improve with //.	Keep your first draft visible. Add // and your upgrade in the same colour.	Did you respond to the check question without erasing your first idea?

Made by Matt

Tier pack · Aspire

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use Box 1 of the arrival drawing (Smooth: 48, 51, 49 cm; Fabric: 27, 29, 28 cm). Calculate the range for each surface.

Help. Subtract the lowest run from the highest run for each row.

2. Why does the non-overlapping span (48–51 cm vs 27–29 cm) give strong evidence for our conclusion?

Help. Compare the 3 cm variation with the 21 cm gap between means.

3. A rover run swerves off the track and hits a book. What should the team do?

Help. Distinguish a method fault from valid repeat variation.

4. State one limitation of applying this classroom table to a real planetary rover.

Help. Compare flat desk strips with planetary terrain.

You do tasks

1. Use Reference A. Analyse the six-run dataset using means, ranges, and non-overlapping spans. Explain why keeping all three repeats is essential. **Staff I ☐ S ☐ C ☐**

Help. Compare the within-condition ranges (3 cm and 2 cm) with the between-condition difference (21 cm).

2. Write a Key Stage 3 conclusion linking the dataset to contact friction. Evaluate how a release gate and side guides improve repeatability. **Staff I ☐ S ☐ C ☐**

Help. Explain how a release gate reduces launch variation.

3. Explain two limitations of using this classroom test as a planetary rover model. Record your upgrade or defence after //. **Staff I ☐ S ☐ C ☐**

Help. Compare flat paper and fabric strips with real planetary ground and rover wheels.

Optional participation-loop prompt. Share your conclusion and improvement with an adult. Hear one feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Why is keeping only the furthest run (51 cm) bad science?

2. How do the non-overlapping spans (48–51 cm vs 27–29 cm) strengthen the friction conclusion?

3. Why can this classroom table not predict a real rover on rough rock?

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Aspire

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt